

HOTEL MANAGEMENT SYSTEM

HOTEL AI SUBSYSTEM & REAL-DATA VERIFICATION REPORT

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1. Executive Summary

A comprehensive end-to-end technical audit was performed across the Hotel AI subsystem spanning the React frontend (hotel-front :5173), Express backend (hotel-back :4000), and FastAPI machine learning microservice (hotel-ai :8000). The investigation confirms that all 5 AI modules are fully operational and actively integrated with real trained machine learning models and the live MySQL production database.

AI Models Verified	FastAPI ML Server	Live MySQL DB	Test Pass Rate
5 Models	Port 8000 (Online)	Connected & Active	100% (9/9 Endpoints)

2. System Architecture & Communication Flow

The Hotel AI system employs a decoupled, microservice architecture optimized for speed, reliability, and graceful degradation:

- Frontend Layer (hotel-front): React 19 + Vite dashboard featuring dedicated AI intelligence tabs, interactive parameter simulators, and a floating operations chatbot.
- Express API Gateway (hotel-back): Node.js/TypeScript gateway handling authentication, role-based access control (RBAC), and direct transactional MySQL queries.
- FastAPI ML Microservice (hotel-ai/backend): High-performance Python 3.11 service loading serialized deep learning and statistical machine learning artifacts into RAM.
- Database Layer: MySQL containing tables for reviews, customers, bookings, rooms, staff, and automated revenue projections.

3. Module-by-Module Technical Assessment

MODULE	AI / ML ENGINE	DATABASE / REAL DATA INTEGRATION	VERIFIED STATUS
1. Review Sentiment Analysis	TensorFlow BiLSTM Neural Net (.keras)	Writes directly to MySQL 'reviews' table; live aggregation on GET /api/reviews	' 100% Live Real Data
2. Operations AI Chatbot	3,200+ Lines Structured SQL Matcher	Live parameterized queries across customers, bookings, rooms & reviews	' 100% Live Real Data
3. Dynamic Revenue Forecast	Random Forest Regressor (.pkl)	Real-time ADR simulation; stores history in 'revenue_predictions'	' 100% Live Real Data
4. Customer Segmentation (RFM)	Scikit-learn K-Means Clustering	Clusters live guest profiles from MySQL 'customers' & 'bookings'	' 100% Live Real Data
5. Booking Cancellation Risk	XGBoost Classifier + Label Encoders	Computes churn probability & monitors active MySQL bookings	' 100% Live Real Data

4. Live Endpoint Diagnostic Test Results

Automated end-to-end integration tests were executed against all live endpoints on ports 8000 and 4000. All services responded with HTTP 200 and verified payloads:

FASTAPI ML MICROSERVICE (http://127.0.0.1:8000)
[PASS] Health Check (GET /) Status 200 OK -> {"message": "Hotel AI API is running"}
[PASS] Sentiment Analysis (POST /api/review/analyze) Status 200 OK -> Input: 'Fabulous room & polite staff' Predicted: 'positive' (Saved to MySQL ID #5)
[PASS] Customer Segmentation (POST /api/customer-segmentation/predict) Status 200 OK -> Input: {stays: 4, spend: 35000} Cluster: 2 (Frequent Corporate)
[PASS] Revenue Forecast (POST /api/revenue-prediction/predict) Status 200 OK -> Input: {lead_time: 20, occupancy: 75%} Predicted: +404.00/night baseline
[PASS] Cancellation Risk (POST /api/booking-cancellation/predict) Status 200 OK -> Input: {lead_time: 45, prev_cancels: 1} Prob: 94.8% (Risk: HIGH)
EXPRESS GATEWAY & DATABASE (http://localhost:4000)
[PASS] Live Reviews Query (GET /api/reviews) Status 200 OK -> Retrieved latest MySQL reviews including newly classified NLP records.
[PASS] Live Segmentation Query (GET /api/customer-segmentation) Status 200 OK -> Aggregated real customer profiles (e.g., Guest 'mohit' Silver/Daily tier).
[PASS] Live Cancellation Query (GET /api/booking-cancellation) Status 200 OK -> Parsed active reservation records and calculated real-time cancellation rate.
[PASS] AI Chatbot SQL Intelligence (POST /api/chatbot) Status 200 OK -> Question: 'How many total guests?' Answer: 'Total customers: 18' (queried from DB)

5. Key Audit Conclusions & Operation Guidelines

- No Mock Data Dependency: The application operates directly with trained AI model weights (.keras, .pkl) and active MySQL records without mock data bypasses.
- Graceful Degradation: All frontend components implement Promise.allSettled and robust fallbacks so the UI remains visually responsive even if the Python service is offline.
- Automated Health Polling: The HotelAiHub component features a continuous 15-second heartbeat poll to ensure staff and administrators have real-time visibility into the Python ML service status.
- Startup Sequence: Ensure all 3 services are active for complete functionality:
 1. Frontend: npm run dev in hotel-front (Port 5173)
 2. Backend: npm run dev in hotel-back (Port 4000)
 3. AI Engine: .\env\Scripts\python.exe -m uvicorn main:app --reload --port 8000 in hotel-ai\backend (Port 8000)

